6. K vecinos mas cercanos

November 4, 2026

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt


data = pd.read_csv("Breast-cancer.csv")
data
```

[1]:

```
           id diagnosis  radius_mean  texture_mean  perimeter_mean  area_mean  \
0      842302         M        17.99         10.38          122.80     1001.0
1      842517         M        20.57         17.77          132.90     1326.0
2    84300903         M        19.69         21.25          130.00     1203.0
3    84348301         M        11.42         20.38           77.58      386.1
4    84358402         M        20.29         14.34          135.10     1297.0
..        …       …          …             …               …         …
564    926424         M        21.56         22.39          142.00     1479.0
565    926682         M        20.13         28.25          131.20     1261.0
566    926954         M        16.60         28.08          108.30      858.1
567    927241         M        20.60         29.33          140.10     1265.0
568     92751         B         7.76         24.54           47.92      181.0

     smoothness_mean  compactness_mean  concavity_mean  concave points_mean  \
0            0.11840           0.27760         0.30010              0.14710
1            0.08474           0.07864         0.08690              0.07017
2            0.10960           0.15990         0.19740              0.12790
3            0.14250           0.28390         0.24140              0.10520
4            0.10030           0.13280         0.19800              0.10430
..               …                 …               …                    …
564          0.11100           0.11590         0.24390              0.13890
565          0.09780           0.10340         0.14400              0.09791
566          0.08455           0.10230         0.09251              0.05302
567          0.11780           0.27700         0.35140              0.15200
568          0.05263           0.04362         0.00000              0.00000

     …  texture_worst  perimeter_worst  area_worst  smoothness_worst  \
0    …          17.33           184.60      2019.0           0.16220
1    …          23.41           158.80      1956.0           0.12380
```

```
2    …          25.53          152.50      1709.0          0.14440
3    …          26.50           98.87       567.7          0.20980
4    …          16.67          152.20      1575.0          0.13740
..   …            …              …           …              …
564  …          26.40          166.10      2027.0          0.14100
565  …          38.25          155.00      1731.0          0.11660
566  …          34.12          126.70      1124.0          0.11390
567  …          39.42          184.60      1821.0          0.16500
568  …          30.37           59.16       268.6          0.08996

     compactness_worst  concavity_worst  concave points_worst  symmetry_worst  \
0              0.66560           0.7119                0.2654          0.4601
1              0.18660           0.2416                0.1860          0.2750
2              0.42450           0.4504                0.2430          0.3613
3              0.86630           0.6869                0.2575          0.6638
4              0.20500           0.4000                0.1625          0.2364
..                 …                …                     …               …
564            0.21130           0.4107                0.2216          0.2060
565            0.19220           0.3215                0.1628          0.2572
566            0.30940           0.3403                0.1418          0.2218
567            0.86810           0.9387                0.2650          0.4087
568            0.06444           0.0000                0.0000          0.2871

     fractal_dimension_worst  Unnamed: 32
0                    0.11890          NaN
1                    0.08902          NaN
2                    0.08758          NaN
3                    0.17300          NaN
4                    0.07678          NaN
..                       …            …
564                  0.07115          NaN
565                  0.06637          NaN
566                  0.07820          NaN
567                  0.12400          NaN
568                  0.07039          NaN

[569 rows x 33 columns]
```

[5]:
```
print(data.columns.tolist())
```

```
['diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean', 'area_mean',
'smoothness_mean', 'compactness_mean', 'concavity_mean', 'concave points_mean',
'symmetry_mean', 'fractal_dimension_mean', 'radius_se', 'texture_se',
'perimeter_se', 'area_se', 'smoothness_se', 'compactness_se', 'concavity_se',
'concave points_se', 'symmetry_se', 'fractal_dimension_se', 'radius_worst',
'texture_worst', 'perimeter_worst', 'area_worst', 'smoothness_worst',
'compactness_worst', 'concavity_worst', 'concave points_worst',
```

```
'symmetry_worst', 'fractal_dimension_worst']
```

[6]:
```
data.drop(["id", "Unnamed: 32"], axis = 1, inplace=True)
data.head()
```

[6]:

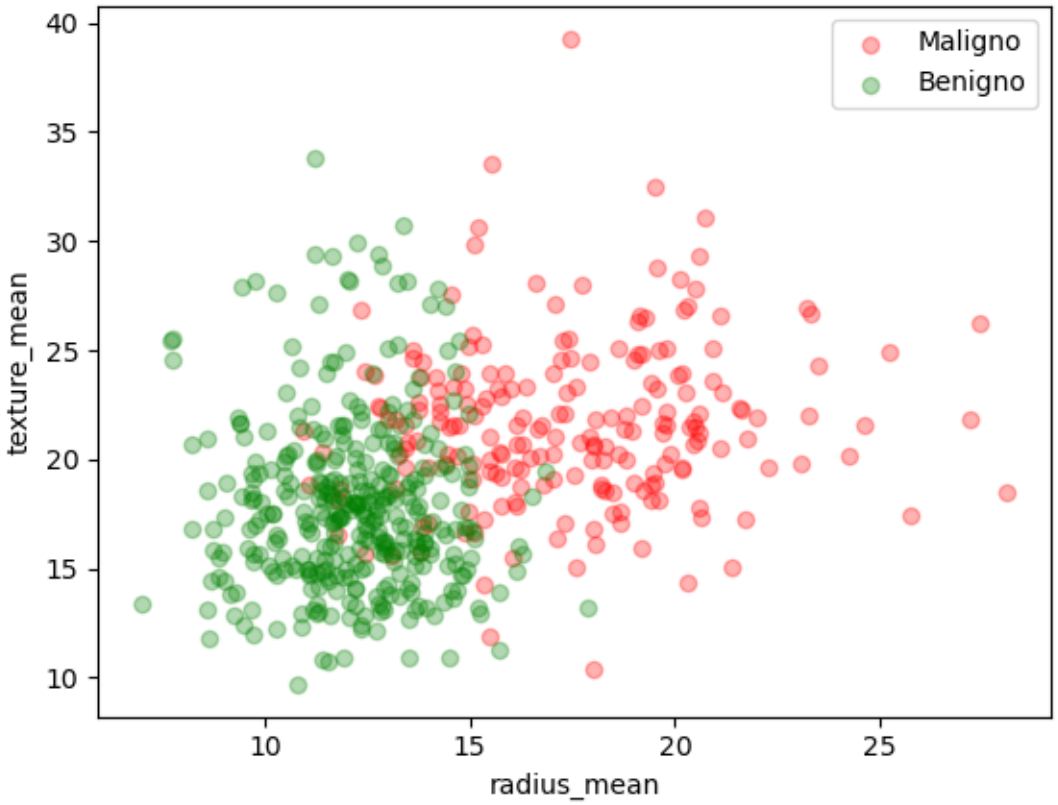
	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	…	radius_worst	texture_worst	perimeter_worst	area_worst	smoothness_worst	compactness_worst	concavity_worst	concave points_worst	symmetry_worst	fractal_dimension_worst
0	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	…	25.38	17.33	184.60	2019.0	0.1622	0.6656	0.7119	0.2654	0.4601	0.11890
1	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	…	24.99	23.41	158.80	1956.0	0.1238	0.1866	0.2416	0.1860	0.2750	0.08902
2	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	…	23.57	25.53	152.50	1709.0	0.1444	0.4245	0.4504	0.2430	0.3613	0.08758
3	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	…	14.91	26.50	98.87	567.7	0.2098	0.8663	0.6869	0.2575	0.6638	0.17300
4	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	…	22.54	16.67	152.20	1575.0	0.1374	0.2050	0.4000	0.1625	0.2364	0.07678

[5 rows x 31 columns]

[7]:
```
M = data[data.diagnosis == "M"]
B = data[data.diagnosis == "B"]

# Diagrama de dispersión
```

```
plt.scatter(M.radius_mean, M.texture_mean, color = "red", label = "Maligno",␣
 ↪alpha = 0.3)
plt.scatter(B.radius_mean, B.texture_mean, color = "green", label = "Benigno",␣
 ↪alpha = 0.3)
plt.xlabel("radius_mean")
plt.ylabel("texture_mean")
plt.legend()
plt.show()
```



[8]:
```
# 0 = Benigno, 1 = Maligno
data.diagnosis = [1 if each == 'M' else 0 for each in data.diagnosis]
data
```

[8]:

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	\
0	1	17.99	10.38	122.80	1001.0	
1	1	20.57	17.77	132.90	1326.0	
2	1	19.69	21.25	130.00	1203.0	
3	1	11.42	20.38	77.58	386.1	
4	1	20.29	14.34	135.10	1297.0	
..	…	…	…	…	…	
564	1	21.56	22.39	142.00	1479.0	

```
565          1        20.13         28.25          131.20     1261.0
566          1        16.60         28.08          108.30      858.1
567          1        20.60         29.33          140.10     1265.0
568          0         7.76         24.54           47.92      181.0

     smoothness_mean  compactness_mean  concavity_mean  concave points_mean  \
0            0.11840           0.27760         0.30010              0.14710
1            0.08474           0.07864         0.08690              0.07017
2            0.10960           0.15990         0.19740              0.12790
3            0.14250           0.28390         0.24140              0.10520
4            0.10030           0.13280         0.19800              0.10430
..               …                 …               …                    …
564          0.11100           0.11590         0.24390              0.13890
565          0.09780           0.10340         0.14400              0.09791
566          0.08455           0.10230         0.09251              0.05302
567          0.11780           0.27700         0.35140              0.15200
568          0.05263           0.04362         0.00000              0.00000

     symmetry_mean  …  radius_worst  texture_worst  perimeter_worst  \
0           0.2419  …        25.380          17.33           184.60
1           0.1812  …        24.990          23.41           158.80
2           0.2069  …        23.570          25.53           152.50
3           0.2597  …        14.910          26.50            98.87
4           0.1809  …        22.540          16.67           152.20
..             …  …             …              …                …
564         0.1726  …        25.450          26.40           166.10
565         0.1752  …        23.690          38.25           155.00
566         0.1590  …        18.980          34.12           126.70
567         0.2397  …        25.740          39.42           184.60
568         0.1587  …         9.456          30.37            59.16

     area_worst  smoothness_worst  compactness_worst  concavity_worst  \
0        2019.0           0.16220            0.66560           0.7119
1        1956.0           0.12380            0.18660           0.2416
2        1709.0           0.14440            0.42450           0.4504
3         567.7           0.20980            0.86630           0.6869
4        1575.0           0.13740            0.20500           0.4000
..          …                 …                  …                …
564      2027.0           0.14100            0.21130           0.4107
565      1731.0           0.11660            0.19220           0.3215
566      1124.0           0.11390            0.30940           0.3403
567      1821.0           0.16500            0.86810           0.9387
568       268.6           0.08996            0.06444           0.0000

     concave points_worst  symmetry_worst  fractal_dimension_worst
0                  0.2654          0.4601                  0.11890
1                  0.1860          0.2750                  0.08902
```

```
2                0.2430          0.3613                0.08758
3                0.2575          0.6638                0.17300
4                0.1625          0.2364                0.07678
..                  …               …                     …
564              0.2216          0.2060                0.07115
565              0.1628          0.2572                0.06637
566              0.1418          0.2218                0.07820
567              0.2650          0.4087                0.12400
568              0.0000          0.2871                0.07039

[569 rows x 31 columns]
```

[9]:
```
y = data.diagnosis.values
x_data = data.drop(["diagnosis"], axis = 1)
```

[10]:
```
y
```

[10]:
```
array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1,
       1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1,
       0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1,
       0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0,
       0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1,
       1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 0,
       0, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1,
       1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 1,
       0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0,
       0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1,
       1, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1,
       1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 1,
       0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0,
       0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1,
       0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0,
       0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0])
```

[11]:
```
x_data
```

[11]:

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	\
0	17.99	10.38	122.80	1001.0	0.11840	
1	20.57	17.77	132.90	1326.0	0.08474	
2	19.69	21.25	130.00	1203.0	0.10960	
3	11.42	20.38	77.58	386.1	0.14250	
4	20.29	14.34	135.10	1297.0	0.10030	
..	…	…	…	…	…	
564	21.56	22.39	142.00	1479.0	0.11100	
565	20.13	28.25	131.20	1261.0	0.09780	
566	16.60	28.08	108.30	858.1	0.08455	
567	20.60	29.33	140.10	1265.0	0.11780	
568	7.76	24.54	47.92	181.0	0.05263	

	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	\
0	0.27760	0.30010	0.14710	0.2419	
1	0.07864	0.08690	0.07017	0.1812	
2	0.15990	0.19740	0.12790	0.2069	
3	0.28390	0.24140	0.10520	0.2597	
4	0.13280	0.19800	0.10430	0.1809	
..	…	…	…	…	
564	0.11590	0.24390	0.13890	0.1726	
565	0.10340	0.14400	0.09791	0.1752	
566	0.10230	0.09251	0.05302	0.1590	
567	0.27700	0.35140	0.15200	0.2397	
568	0.04362	0.00000	0.00000	0.1587	

	fractal_dimension_mean	…	radius_worst	texture_worst	\
0	0.07871	…	25.380	17.33	
1	0.05667	…	24.990	23.41	
2	0.05999	…	23.570	25.53	
3	0.09744	…	14.910	26.50	
4	0.05883	…	22.540	16.67	
..	…	…	…	…	
564	0.05623	…	25.450	26.40	
565	0.05533	…	23.690	38.25	
566	0.05648	…	18.980	34.12	
567	0.07016	…	25.740	39.42	
568	0.05884	…	9.456	30.37	

	perimeter_worst	area_worst	smoothness_worst	compactness_worst	\
0	184.60	2019.0	0.16220	0.66560	
1	158.80	1956.0	0.12380	0.18660	
2	152.50	1709.0	0.14440	0.42450	
3	98.87	567.7	0.20980	0.86630	
4	152.20	1575.0	0.13740	0.20500	
..	…	…	…	…	
564	166.10	2027.0	0.14100	0.21130	

```
565          155.00      1731.0          0.11660           0.19220
566          126.70      1124.0          0.11390           0.30940
567          184.60      1821.0          0.16500           0.86810
568           59.16       268.6          0.08996           0.06444

     concavity_worst  concave points_worst  symmetry_worst  \
0             0.7119                0.2654          0.4601
1             0.2416                0.1860          0.2750
2             0.4504                0.2430          0.3613
3             0.6869                0.2575          0.6638
4             0.4000                0.1625          0.2364
..               …                     …               …
564           0.4107                0.2216          0.2060
565           0.3215                0.1628          0.2572
566           0.3403                0.1418          0.2218
567           0.9387                0.2650          0.4087
568           0.0000                0.0000          0.2871

     fractal_dimension_worst
0                    0.11890
1                    0.08902
2                    0.08758
3                    0.17300
4                    0.07678
..                       …
564                  0.07115
565                  0.06637
566                  0.07820
567                  0.12400
568                  0.07039

[569 rows x 30 columns]
```

[12]:
```
# Normalización
x = (x_data - np.min(x_data)) / (np.max(x_data) - np.min(x_data))
x
```

[12]:
```
     radius_mean  texture_mean  perimeter_mean  area_mean  smoothness_mean  \
0       0.004229      0.002440        0.028867   0.235308         0.000028
1       0.004835      0.004177        0.031241   0.311707         0.000020
2       0.004629      0.004995        0.030559   0.282793         0.000026
3       0.002685      0.004791        0.018237   0.090762         0.000033
4       0.004770      0.003371        0.031758   0.304890         0.000024
..           …             …               …          …                …
564     0.005068      0.005263        0.033380   0.347673         0.000026
565     0.004732      0.006641        0.030842   0.296427         0.000023
566     0.003902      0.006601        0.025458   0.201716         0.000020
```

```
567     0.004843       0.006895         0.032934   0.297367          0.000028
568     0.001824       0.005769         0.011265   0.042548          0.000012

     compactness_mean  concavity_mean  concave points_mean  symmetry_mean  \
0            0.000065        0.000071             0.000035       0.000057
1            0.000018        0.000020             0.000016       0.000043
2            0.000038        0.000046             0.000030       0.000049
3            0.000067        0.000057             0.000025       0.000061
4            0.000031        0.000047             0.000025       0.000043
..                …               …                    …              …
564          0.000027        0.000057             0.000033       0.000041
565          0.000024        0.000034             0.000023       0.000041
566          0.000024        0.000022             0.000012       0.000037
567          0.000065        0.000083             0.000036       0.000056
568          0.000010        0.000000             0.000000       0.000037

     fractal_dimension_mean  …  radius_worst  texture_worst  \
0                  0.000019  …      0.005966       0.004074
1                  0.000013  …      0.005874       0.005503
2                  0.000014  …      0.005541       0.006001
3                  0.000023  …      0.003505       0.006229
4                  0.000014  …      0.005299       0.003919
..                      …  …           …              …
564                0.000013  …      0.005983       0.006206
565                0.000013  …      0.005569       0.008992
566                0.000013  …      0.004462       0.008021
567                0.000016  …      0.006051       0.009267
568                0.000014  …      0.002223       0.007139

     perimeter_worst  area_worst  smoothness_worst  compactness_worst  \
0           0.043394    0.474612          0.000038           0.000156
1           0.037330    0.459803          0.000029           0.000044
2           0.035849    0.401740          0.000034           0.000100
3           0.023242    0.133451          0.000049           0.000204
4           0.035778    0.370240          0.000032           0.000048
..               …           …                 …                  …
564         0.039046    0.476493          0.000033           0.000050
565         0.036436    0.406911          0.000027           0.000045
566         0.029784    0.264222          0.000027           0.000073
567         0.043394    0.428068          0.000039           0.000204
568         0.013907    0.063141          0.000021           0.000015

     concavity_worst  concave points_worst  symmetry_worst  \
0           0.000167              0.000062        0.000108
1           0.000057              0.000044        0.000065
2           0.000106              0.000057        0.000085
3           0.000161              0.000061        0.000156
```

```
4           0.000094              0.000038        0.000056
..               …                     …             …
564         0.000097              0.000052        0.000048
565         0.000076              0.000038        0.000060
566         0.000080              0.000033        0.000052
567         0.000221              0.000062        0.000096
568         0.000000              0.000000        0.000067

     fractal_dimension_worst
0                   0.000028
1                   0.000021
2                   0.000021
3                   0.000041
4                   0.000018
..                       …
564                 0.000017
565                 0.000016
566                 0.000018
567                 0.000029
568                 0.000017

[569 rows x 30 columns]
```

[13]:
```
# Bases de entrenamiento y prueba
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.30,␣
 ↪random_state=1)
```

[14]:
```
# Bases de entrenamiento y prueba
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.30,␣
 ↪random_state=1)
```

[16]:
```
# Modelo KNN (K vecinos más cercanos)
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(x_train, y_train)
prediction = knn.predict(x_test)
print("Score: ", knn.score(x_test, y_test))
```

```
Score:  0.9239766081871345
```

[17]:
```
y_pred = prediction
y_true = y_test
```

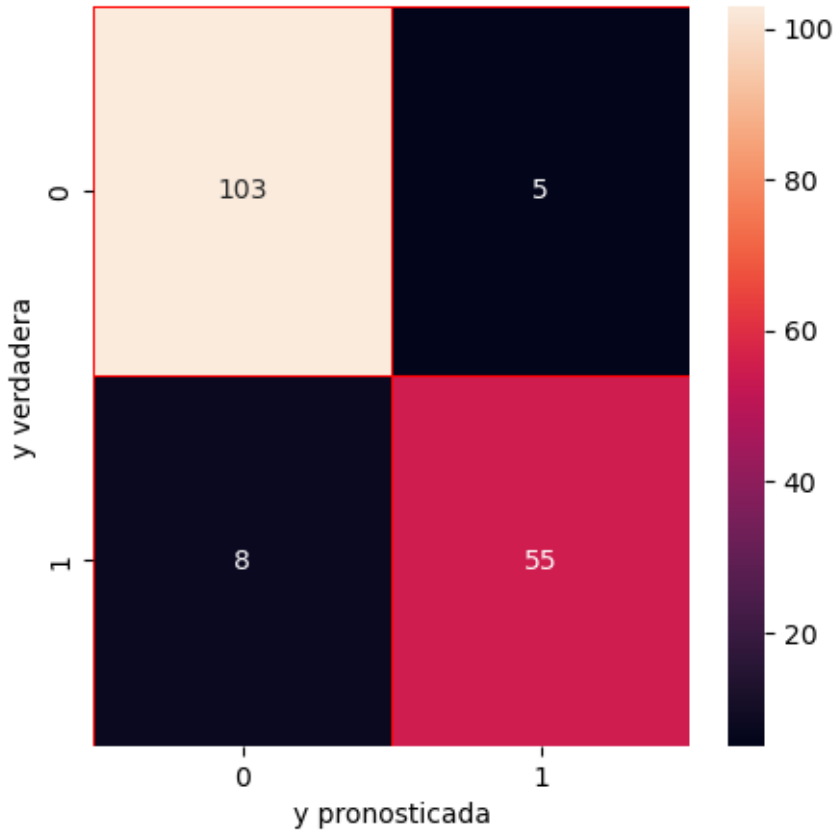
[18]:
```
# Creación de la matriz de confusión
from sklearn.metrics import confusion_matrix
```

```
cm = confusion_matrix(y_true, y_pred)
cm
```

[18]:
```
array([[103,   5],
       [  8,  55]])
```

[19]:
```
# Visualización de la matriz de confusión
import seaborn as sns
f, ax = plt.subplots(figsize = (5,5))

sns.heatmap(cm, annot = True, linewidths = 0.5, linecolor = "red", fmt = ".0f",␣
 ↪ax = ax)
plt.xlabel("y pronosticada")
plt.ylabel("y verdadera")
plt.show()
```



[20]:
```
# Cálculo de la precisión global
Correctos = cm[0,0] + cm[1,1]
Incorrectos = cm[0,1] + cm[1,0]
PrecisionGlobal = Correctos / (Correctos + Incorrectos)
```

```
PrecisionGlobal
```

[20]: 0.9239766081871345

[21]:
```
PrecisionBenigno = cm[0,0] / (cm[0,0] + cm[1,0])
PrecisionBenigno
```

[21]: 0.9279279279279279

[22]:
```
PrecisionMaligno = cm[1,1] / (cm[1,1] + cm[0,1])
PrecisionMaligno
```

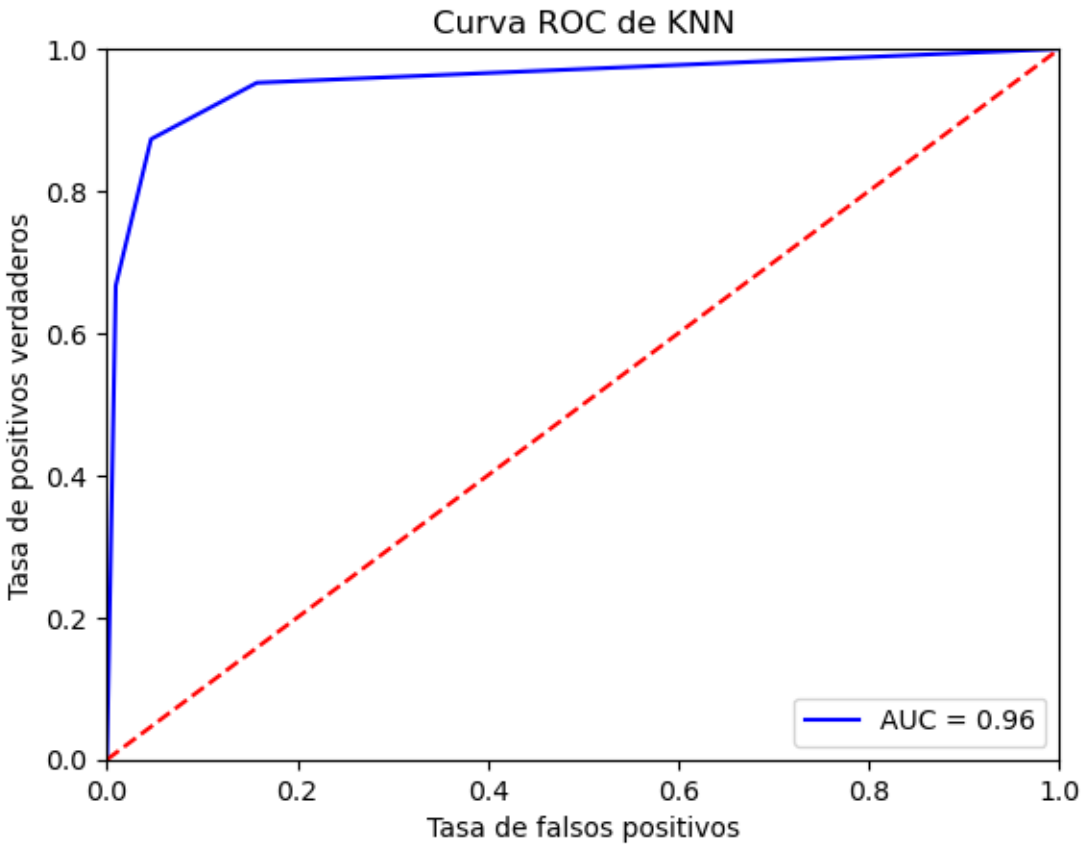
[22]: 0.9166666666666666

[24]:
```
from sklearn.metrics import roc_curve
from sklearn.metrics import auc

y_scores = knn.predict_proba(x_test)

fpr, tpr, threshold = roc_curve(y_test, y_scores[:,1])
# fpr = False positive rate, tpr = True positive rate
roc_auc = auc(fpr, tpr)

plt.title('Receiver Operating Characteristic')
plt.plot(fpr, tpr, 'b', label = 'AUC = %0.2f' % roc_auc)
plt.legend(loc = 'lower right')
plt.plot([0,1], [0,1], 'r--')
plt.xlim([0,1])
plt.ylim([0,1])
plt.ylabel("Tasa de positivos verdaderos")
plt.xlabel("Tasa de falsos positivos")
plt.title("Curva ROC de KNN")
plt.show()
```


Curva ROC de KNN
Tasa de positivos verdaderos
Tasa de falsos positivos
AUC = 0.96
0.0
0.2
0.4
0.6
0.8
1.0

[ ]: